Cancer Classification Report

1. Dataset Summary

Parameter	Value
Total Samples	760
Cancer Samples	414
Healthy Samples	346
Raw TSS Sites	28,121
After Expression Filter (≥10 samples)	24,541
After Variance Filter (Top 2000)	2,000
Final Feature Matrix	760 × 2,000

2. PCA Summary

Component	Variance Explained
PC1	0.288 (28.8%)
PC2	0.026 (2.6%)
PC3	0.020 (2.0%)
Total — 100 Components	0.618 (61.8%)

3. Classifier Performance

Metric	Random Forest	Gradient Boosting	Logistic Regression
Input Features	2,000 TSS	100 PCA	100 PCA
5-Fold ROC-AUC	0.863 ± 0.008	0.851 ± 0.018	0.839 ± 0.026
5-Fold Accuracy	0.772 ± 0.016	0.768 ± 0.017	0.749 ± 0.024
Hold-out Accuracy	0.7526	0.7947	0.7737
Hold-out ROC-AUC	0.8593	0.8741	0.8717

4. Confusion Matrices (Hold-out Set, n = 190)

Random Forest

	Predicted Healthy	Predicted Cancer
Actual Healthy (n=87)	60	27
Actual Cancer (n=103)	20	83

Gradient Boosting

	Predicted Healthy	Predicted Cancer
Actual Healthy (n=87)	66	21

Actual Cancer (n=103)	18	85

Logistic Regression

	Predicted Healthy	Predicted Cancer
Actual Healthy (n=87)	68	19
Actual Cancer (n=103)	24	79

5. Classification Reports

Random Forest

Class	Precision	Recall	F1-Score	Support
Healthy (0)	0.75	0.69	0.72	87
Cancer (1)	0.75	0.81	0.78	103
Macro Avg	0.75	0.75	0.75	190
Weighted Avg	0.75	0.75	0.75	190

Gradient Boosting

Class	Precision	Recall	F1-Score	Support
Healthy (0)	0.79	0.76	0.77	87
Cancer (1)	0.80	0.83	0.81	103
Macro Avg	0.79	0.79	0.79	190
Weighted Avg	0.79	0.79	0.79	190

Logistic Regression

Class	Precision	Recall	F1-Score	Support
Healthy (0)	0.74	0.78	0.76	87
Cancer (1)	0.81	0.77	0.79	103
Macro Avg	0.77	0.77	0.77	190
Weighted Avg	0.78	0.77	0.78	190

6. Top 30 Discriminative TSS Sites (Random Forest)

Rank	Gene	Importance
1	HPN-AS1	0.008067
2	MBD3L2	0.007127
3	GP1BB	0.007049
4	MBD3L3	0.006129
5	DUSP5P1	0.005123
6	MIR6859-2	0.005045

Rank	Gene	Importance
7	MIR6859-1	0.004833
8	F2RL3	0.004817
9	MIR6859-3	0.004461
10	MIR6859-4	0.004305
11	RNF183	0.004283
12	LOC101929657	0.004267
13	LOC101929141	0.003790
14	LOC100507156	0.003527
15	NEK5	0.003500
16	MIR4441	0.003393
17	ZSWIM2	0.003189
18	RNA5S4	0.003015
19	RNA5S14	0.002903
20	PTGIR	0.002880
21	TRPS1	0.002875
22	ADCY3	0.002811
23	CXADRP2	0.002616
24	RNA5S8	0.002589
25	ZNF654	0.002584
26	LINC00689	0.002567
27	CLEC4C	0.002559
28	LOC101929538	0.002532
29	H1-7	0.002507
30	LOC102503427	0.002505

7. Top 5 Genes — Cancer vs Healthy TSS Coverage

Gene	Cancer Mean	Healthy Mean	Fold Ratio	Direction
HPN-AS1	0.0356	0.0595	0.60x	Depleted in Cancer
MBD3L2	0.0411	0.0796	0.52x	Depleted in Cancer
GP1BB	0.0853	0.1349	0.63x	Depleted in Cancer
MBD3L3	0.0353	0.0685	0.51x	Depleted in Cancer
DUSP5P1	0.0599	0.0304	1.97x	Enriched in Cancer